

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Bachelor of Technology (Computer Science and Engineering) SEMESTER - 3 Winter 2025 (Regular)
Course : Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology
Semester : SEMESTER - 3
Subject Code & Name: 25AF12451PC304 - OBJECT-ORIENTED PROGRAMMING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 In a Java program, the heap is used for:

- a) Storing method call details
- b) Storing local variables of methods
- c) Storing dynamically created objects
- d) Storing arithmetic results

A Car class has instance variables model and speed. If multiple objects of Car are created, what happens to these variables?

- a) Shared among all objects
- b) Each object gets its own copy
- c) They are automatically set to default values of null
- d) They must be declared static

3 If an object is declared but not initialized, what will its reference contain?

- a) Default constructor values
- b) Zero
- c) Null
- d) Garbage values

What will be the output of the following Code

```
public class Test {  
    public static void main(String[] args) {  
        int a = 5/2;  
        double b = 5/2;  
        double c = 5/2.0;  
        System.out.println(a + " " + b + " " + c);  
    }  
}
```

- a) 2 2.0 2.5
- b) 2 2.5 2.5
- c) 2 2.0 2.0
- d) Error

- 5 What is method overloading in Java OOPs?
 a) Having two methods with the same name but different return types
 b) Having two methods with the same name and parameters but different return types
 c) Having two methods with the same name and parameters but different implementation
 d) Having two methods with the same name but different parameters
- 6 What is a static method in Java OOPs?
 a) A method that can be accessed without creating an object of the class
 b) A method that is automatically invoked when an object is created
 c) A method that is used to override other methods
 d) A method that is used to create objects
- What is the correct syntax for declaring a public method named 'display' in Java?
 a) public void display()
 b) private void display()
 c) protected void display()
 d) void display()
- 8 Which access modifier restricts access of a class member to only its subclasses and classes in the same package? <https://www.pyqonline.com>
 a) Private
 b) Default
 c) Public
 d) Protected
- Which keyword is used to inherit a class in Java?
 a) implement
 b) inherit
 c) extends
 d) super
- 10 Which keyword is used to explicitly generate an exception?
 a) throws
 b) catch
 c) throw
 d) try
- 11 Which package contains Scanner class?
 a) java.io
 b) java.util
 c) java.lang
 d) java.ne
- Which feature of Java supports modularization?
 a) Inheritance
 b) Packages
 c) Polymorphism
 d) Encapsulation

Q2. Solve the following.

- A) A company needs a *Employee* class to store *name*, *ID*, and *salary*. Write a Java program to create objects of this class and display employee details. 6
- B) Explain how objects are initialized using constructors in Java. Describe the use of setter and getter methods. Also, explain the difference between primitive and reference data types. 6

Q3. Solve the following.

- A) Explain modularization in Java. Describe Java packages, importing packages with a suitable example. 6
- B) Develop a Java program to demonstrate constructor overloading by creating multiple constructors in a class. 6

Q4. Solve Any Two of the following.

- A) Develop a Java program to demonstrate the use of final methods and final classes. 6
- B) Implement a program using interfaces to achieve multiple inheritance. 6
- C) Write a Java program to demonstrate runtime polymorphism using method overriding. 6

Q5. Solve Any Two of the following.

- A) Write a Java program to handle *ArithmeticException* and *InputMismatchException* using try-catch-finally blocks. 6
- B) Develop a program using Iterator to traverse a collection safely. 6
- C) Develop a program that demonstrates multiple catch blocks for handling different exceptions. 6

Q6. Solve Any Two of the following.

- A) Create a Java program that uses both static and non-static members and explain their interaction. 6
- B) Design a Java program that demonstrates the Java exception hierarchy using custom exceptions. 6
- C) Write a Java program using an abstract class and its subclasses to show polymorphic behavior. 6

*** End ***